ModelIt!

Student Activity Pack

Ecohydrology — Fire-Resistant Landscapes

How Beavers Create Nature's Firebreaks

Grade NE | NGSS MS-ESS2-4, MS-ESS3-3, MS-ESS3-4, MS-LS2-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **In Level 1 Sponge vs. Pavement, you learned that ground type affects water absorption and runoff. What happens to water absorption in a NATURAL landscape during drought?**

2. **Have you ever seen areas that stayed green when everything around them was brown and dry? Why do you think that happens?**

3. **How might a beaver dam affect the land AROUND it — not just the pond behind it?**

4. **Draw what you think happens underground when a beaver dam stores water.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Precipitation | Beaver Dam Count | Drought Severity | Groundwater Level | Soil Moisture | Vegetation Health | Surface Runoff | Fire Spread Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: With Beavers in Drought

Many dams, low rain, high drought

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Without Beavers in Drought

No dams, low rain, high drought

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Normal Rainfall

Moderate dams, normal conditions

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Speed — How fast the wind blows during a fire event — wind accelerates fire spread and dries vegetation faster

Terrain Slope — How steep the land is — fire moves faster uphill and water drains faster on steep slopes

Vegetation Density — How thick the plant cover is — denser vegetation can mean more fuel OR more moisture depending on health

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do beaver dams create fire-resistant green corridors in landscapes devastated by wildfire?
Use evidence from your simulation to support your answers.

1. **How is the Level 2 ecohydrology model different from the Level 1 Sponge vs. Pavement model? What new components reveal about fire resistance?**

2. **Compare the With Beavers and Without Beavers drought scenarios — what is the CRITICAL difference in the cascade?**

3. **Why are beavers sometimes called 'nature's firefighters' even though they do not fight fire directly?**

4. **What surprised you most when comparing the simulation scenarios?**

5. **How could land managers use this model to make decisions about wildfire prevention?**

8 STEM Challenge: Drought vs. Dam Time-Lapse Terrarium

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YOUR ENGINEERING MISSION

Build two terrariums — one with a miniature dam and water reservoir, one without — then withhold water for two weeks to observe which landscape stays green longer.

THE SCENARIO

A wildfire research team wants to demonstrate how beaver dams protect landscapes during drought. Your team will build a physical model that shows the difference between a dammed and undammed watershed over a simulated drought period.

Engineering Design Thinking

1. HOW WILL YOU BUILD A MINIATURE DAM AND RESERVOIR THAT ACTUALLY STORES WATER IN THE SOIL?
2. WHAT PLANTS WILL YOU USE THAT WILL SHOW VISIBLE CHANGES WHEN THEY LOSE WATER?
3. HOW WILL YOU MEASURE SOIL MOISTURE AT DIFFERENT DEPTHS OVER THE TWO WEEKS?
4. HOW WILL YOU CONTROL ALL VARIABLES SO THE ONLY DIFFERENCE BETWEEN THE TWO TERRARIUMS IS THE DAM?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Ecohydrologists study how water and ecosystems interact. They work on wildfire management, watershed restoration, and climate adaptation projects for government agencies, universities, and conservation organizations. They earn $60,000–$100,000/year.